Q1

a) Def ~~Wes~~ For integers a, b we say a divides b, written as $a|b$ iff $\exists\, c \in \mathbb{Z}$ such that $b = ac$.

To prove transitivity we assume.

$a|b$ and $b|c$.

ie. $\exists\ \underline{\alpha, \beta \in \mathbb{Z}}$ such that

$b = \alpha a$ and $c = \beta b$.

$\Rightarrow c = \beta b = \beta \alpha a$

$\Rightarrow c = \beta\alpha a$, and clearly $\beta\alpha \in \mathbb{Z}$

$\Rightarrow a|c$

~~gcd(0,0)~~ not both zero.

b) Definition ~~The g~~ For integers a, b $\wedge$ the $\gcd(a,b)$ is defined as the greatest common divisor of a & b, ie. the largest integer d satisfying $d|a$ and $d|b$.

If $d = \gcd(a,b)$ then d is the smallest positive linear combination of a, b.

ie. the smallest positive integer of the form $ma + nb$, for $m, n \in \mathbb{Z}$.

and moreover every linear combination $ma + nb$ is a multiple of d.

c) We move through a sequence of integer divisions with remainder, noting the associated equations between the relevant gcds

$$\begin{array}{r} 246 \\ 178 \\ \hline 68 \end{array}$$

$$246 = 2 \cdot 89 + 68 \Rightarrow \gcd(246, 89) = \gcd(89, 68)$$

$$89 = 1 \cdot 68 + 21 \Rightarrow \gcd(89, 68) = \gcd(68, 21).$$

$$68 = 3 \cdot 21 + 5 \Rightarrow = \gcd(21, 5)$$

$$21 = 4 \cdot 5 + 1 \Rightarrow = \gcd(5, 1)$$

$$5 = 5 \cdot 1 + 0 \Rightarrow = \gcd(1, 0) = 1.$$

d). Claim: $\forall n \geq 1 \quad 13 \mid 9^{3n} - 1$.

Using induction we first check the base case. $n = 1$

$$\begin{array}{r} 81 \\ 9 \\ \hline 729 \end{array}$$

$$9^{3 \cdot 1} - 1 = 729 - 1 = 728$$
$$= 56 \cdot 13$$

So the base case is true, ie $13 \mid 9^3 - 1$

Next, assume the result is true for some $k \geq 1$.

ie. $13 \mid 9^{3k} - 1$

Now we try to deduce from this the truth of the ~~rest~~ result for $k+1$.

$$9^{3(k+1)} - 1 = 9^{3k+3} - 1$$

$$= 9^3 \cdot 9^{3k} - 1$$

$$= 9^3 (9^{3k} - 1) + 9^3 - 1$$

$$= 9^3 (9^{3k} - 1) + 56 \cdot 13$$

This is a linear combination of $q^{3k}-1$ and 13, both of which are divisible by 13. Therefore

$$13 \mid q^{3(k+1)}-1.$$

So by the principle of induction, for all $n \geq 1$

$$13 \mid q^{3n}-1.$$

Q2

b). Claim: p^2-1 is never prime for primes $p>2$.

Let p be a prime $p>2$.

Then dividing p by 8 will give. $\cancel{0}, 1, \cancel{2}, 3, \cancel{4}, 5, \cancel{6}, 7$

$$p = 8q + r$$

where r is one of $\{1, 3, 5, 7\}$.

Examining each case:

$r=1$ i.e. $p = 8q+1$, $q \geq 1$.

$$p^2-1 = (8q+1)^2-1 = 64q^2+16q$$
$$= 16\cdot(4q^2+q)$$

which is clearly composite.

$\underline{r=3}$ $p^2-1 = (8q+3)^2 - 1$

$$= 64q^2 + 48q + 8$$

$$= 8(8q^2 + 6q + 1)$$

again composite.

$\underline{r=7}$ $p^2-1 = (8q+7)^2 - 1$

$$= 64q^2 + 112q + 48$$

$$= 2(32q^2 + 56q + 24)$$

again a composite.

$\underline{r=5}$ $p^2-1 = (8q+5)^2 - 1$

$$= 64q^2 + 80q + 24$$

$$= 2(32q^2 + 40q + 12)$$

clearly a composite.

So p^2-1, when ~~$p>2$~~ p is prime and >2 is never a prime.

$$(p^2-1) = (p+1)(p-1).$$

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c). Euclid's lemma.

For a prime p and integers a, b.

$$p \mid ab \implies (p \mid a \text{ or } p \mid b)$$

<u>Proof</u>. Assume $\boxed{p \mid ab.}$

i.e. $ab = cp$.

In addition assume $p \nmid a$

$\implies \gcd(a, p) = 1$

$\implies \exists\ m, n \in \mathbb{Z}$ such that

$ma + np = 1$, Bezout's identity.

$\implies mab + npb = b$.

$\implies mcp + npb = b$

$\implies b = (mc + nb)p$

so $p \mid b$ as required.

This proves that

$$p \mid ab \implies (p \nmid a \implies p \mid b)$$

which is equivalent to

$$p \mid ab \implies (p \mid a \ \underline{\text{OR}}\ p \mid b).$$